

Aaron Beavers — Resume (FutureSight / AI Knowledge Platform)

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AI Infrastructure Engineer (Knowledge Platforms · RAG · Multi-Tenant Systems)

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GitHub: <https://github.com/riatzukiza> · Upwork: <https://www.upwork.com/freelancers/~01d661364c3f86022f>

Summary

AI infrastructure engineer specializing in multi-tenant knowledge platforms: RAG pipelines, vector search, HITL labeling, and training data workflows. 10+ years building production systems pre-agentic-AI; now use agentic tooling to accelerate delivery while keeping changes verifiable. Designed and built the architecture for a domain-aware knowledge management platform spanning public-facing assistants, agent-aware CMS, and expert review workbenches—with tiered embedding strategies (nomic/qwen3) and semantic compaction over event lakes.

Skills

- **AI/ML infrastructure:** RAG pipelines, vector search (ChromaDB, MongoDB Vector Search), tiered embedding strategies (nomic-embed-text, qwen3-embedding), semantic compaction, dual-tier RRF search, HITL labeling workflows

- **Knowledge platforms:** Multi-tenant data isolation, document lifecycle management (draft/review/public/archive), content curation CMS, training data export (SFT JSONL, RLHF preference pairs)
- **Full stack:** TypeScript/Node.js (Fastify), Python (FastAPI), React, Clojure
- **Data:** DuckDB, MongoDB 8.2 (with \$vectorSearch), PostgreSQL, Qdrant, ChromaDB
- **DevOps:** Docker/Compose, Nginx, Prometheus/Grafana observability, multi-arch OCI builds
- **Security:** Multi-tenant isolation, PII detection/handling, audit trails, Bearer token auth, retrieval-native access control

Recent Work (AI Knowledge Platforms)

OpenPlanner — API-First Event Data Lake

2025 – Present - Built a local-first event data lake with DuckDB for structured storage, ChromaDB for vector search, and content-addressed blob storage. Ingests events from ChatGPT exports, Discord, OpenCode sessions, and live social media firehose (Bluesky/AT Protocol). - Implemented **dual-tier semantic memory**: hot collection (fast ingest, nomic/qwen3-0.6b embeddings) + compact collection (semantic packs built by compaction job, qwen3-8b embeddings) with RRF (reciprocal rank fusion) search merging across both tiers. - Extended with **document-oriented endpoints** for the knowledge management platform: document CRUD, warm-tier embedding (qwen3-4b), chunked storage, and visibility-aware search (public/internal/review/archived). - Added **per-source/kind embedding model resolution** with override precedence (project → source → kind → default), Prometheus metrics, Grafana dashboards, and Bearer token auth. <https://github.com/riatzukiza/devel> (services/openplanner/)

Shibboleth — HITL Labeling DSL + Training Data Pipeline

2024 – Present - Authored a generative Clojure DSL for building multilingual adversarial prompt evaluation datasets with deterministic seeds, full provenance, leakage-proof splits (sentence-transformers + HDBSCAN), and SHA-256 manifests. - Extended with a **corporate QA labeling schema** (8 label dimensions: correctness, groundedness, completeness, tone, risk, PII leakage, translation quality, overall label) and a React-based expert review UI for human-in-the-loop data curation. - Pipeline produces SFT datasets and RLHF preference pairs

from labeled examples, with Shibboleth-style datasheets and reproducibility bundles. <https://github.com/octave-commons/shibboleth>

Open Hax Proxy — Multi-Provider LLM Gateway

2023 – Present - Built an OpenAI-compatible gateway integrating multiple LLM providers (OpenAI Responses, Anthropic Messages, Ollama) via model-aware routing, OAuth-based account ingestion, and rate-limit-aware rotation/fallback. - Shipped a React/Vite console with Chroma-backed semantic session search, reasoning-trace preservation, and optional OTEL export. <https://github.com/open-hax/proxx>

Futuresight KMS — Multi-Tenant Knowledge Management Platform

2026 - Designed a **five-layer architecture**: Public Assistant (chat widget) → Agent-Aware CMS (knowledge curation) → Knowledge Worker (internal RAG) → Coding Agent (dev tools) → SME Review (Shibboleth-powered labeling). - Spec'd the **CMS boundary layer** with document visibility states (internal → review → public → archived), AI-assisted draft generation, and publish flow that syncs curated content to a public-facing vector collection. - Defined multi-tenant control plane: tenant isolation, retrieval-native access control, per-tenant encryption, cascading offboarding. - Designed **MongoDB Vector Search unification**: replacing DuckDB + ChromaDB with a single MongoDB 8.2 instance using \$vectorSearch, \$search, and \$searchMeta with mongot sidecar. - Built Zod/Pydantic label schemas, TypeScript API bridge clients, Python FastAPI label CRUD with SFT/RLHF export, and React knowledge labeler component.

Prometheus / Grafana Observability Stack

2025 – Present - Designed and deployed observability for the full data lake: Prometheus metrics (event ingest rate, search latency, compaction job duration, embedding queue depth), Grafana dashboards for data lake health, and alerting for pipeline failures.

Experience

Raft Technologies — Full Stack Engineer (Python/Django + React)

Jul 2020 – Sep 2023 - Built and maintained Celery-based pipelines for uploaded-file processing (parse → validate → summarize → report → notify) designed for safe failure modes and operational visibility. - Integrated anti-malware scanning automation (ClamAV REST) using retry/backoff + timeouts with persisted outcomes for auditability; implemented robust file type + encoding detection. - Supported end-to-end browser automation practices (Cypress E2E + CI execution) to prevent regressions.

CloudApp — Full Stack Engineer (Ruby on Rails)

Sep 2018 – Feb 2019 - Integrated a proprietary OCR service into a screenshot platform; implemented interactive OCR UX (bounding boxes + click-to-extract text) and hardened key flows with automated tests.

Additional Experience

1brand (Full Stack Engineer, 2019–2020) · Birdseed (Full Stack Engineer, 2019) · Freelance Software Engineer (2014–Present)

Education

Valley Springs High School — Diploma (2012)

Work Authorization

Authorized to work in the U.S. for any employer · Fully remote (U.S.) · Open to international contracting

```
fnord:v1 proof=gh(open-hax/proxx octave-commons/shibboleth)
caps=rag(dual-tier hot-warm-compact rrf fusion) embed(nomic
qwen3-0.6b qwen3-4b qwen3-8b) hitl(8-dim-labeling sft-rlhf-export)
lake(duckdb chromadb mongodb-vector-search mongot) cms(visibility-
draft-publish-archive) multi-tenant(retrieval-native pii-detection
```

```
audit-trail) ops(docker compose prometheus grafana fastify  
fastapi)
```